

Research Paper 04 Summary

Methods for Global Survey of Natural Gas Flaring from VIIRS Data

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Elvidge et al. (2016), *Energies*

1 Basic Information

- **Paper Title:** Methods for Global Survey of Natural Gas Flaring from VIIRS Data
- **Authors:** Christopher D. Elvidge, Mikhail Zhizhin, Kimberly Baugh, Feng-Chi Hsu, and Tilottama Ghosh (2016)
- **Core Contribution:** Developed the **VIIRS Nightfire (VNF)** algorithm to detect and calculate the exact temperatures and surface areas of industrial gas flares globally.

2 What is this Paper About? (In Simple Words)

This is one of the most important papers for our project. The authors at NOAA discovered a way to measure the exact temperature and physical size of fires from space at night.

- **The Nighttime Advantage:** During the day, sunlight bouncing off the ground creates noise. At night, there is no sunlight, making any hot fire or flare stand out clearly against the pitch-black Earth.
- **Multi-Band Color Curve (Planck Fitting):** Instead of checking just one infrared camera channel, the algorithm looks at multiple spectral channels (Shortwave, Near-Infrared, and Mid-Infrared). By fitting a physics curve across these channels, it calculates two exact numbers for every hot spot:
 1. **Actual Flame Temperature** (T in Kelvin)
 2. **Effective Surface Area** (S in square meters)

3 Key Discovery: The Temperature Ranges of Different Fires

The paper provides the definitive temperature chart separating industrial heat from natural fires:

4 How This Helps Our Project (PS26162)

4.1 1. Defining the Exact Class Boundaries for Our AI Model

The temperature and area ranges from this paper give our machine learning classifier clear rules for categorizing detections:

Combustion Type	Temperature Range (K)	Typical Physical Footprint
Natural Gas Flaring	1200 K – 1800 K	Very small ($< 50 \text{ m}^2$)
Industrial Furnaces / Smelting	900 K – 1400 K	Compact ($50 \text{ m}^2 - 200 \text{ m}^2$)
Wildfires / Forest Fires	600 K – 900 K	Sprawling (Hundreds of m^2)
Agricultural Crop Burning	500 K – 800 K	Wide, shallow fields

- If a thermal point has $T > 1200 \text{ K}$ and a small source area, it is almost certainly an **industrial gas flare**.
- If it has T between 900 K and 1200 K, it matches **steel mills or industrial furnaces**.
- If $T < 900 \text{ K}$ with a large spatial footprint, it is a **wildfire or agricultural burn**.

4.2 2. Feature Engineering: Temperature vs. Area Ratio

Instead of relying only on raw Fire Radiative Power (FRP), we can engineer a crucial discriminator feature:

$$\text{Intensity Ratio} = \frac{\text{Calculated Temperature}}{\text{Effective Emission Area}} \quad (1)$$

Industrial flares have a very high ratio (huge heat concentrated in a tiny area), whereas forest fires have a low ratio (moderate heat spread across a wide zone).

4.3 3. Building a Reliable Baseline of Indian Refineries and Gas Fields

Using nighttime VIIRS data following Elvidge’s methodology allows us to automatically map and catalog all persistent refinery flare stacks across India (such as Reliance Jamnagar, ONGC offshore rigs, and IOCL refineries) without daytime solar interference.